Reliable Smartphone-Based Detection of Counterfeit Bangladeshi Banknotes: Multi-View Deep Learning with Out-of-Distribution Rejection and a Benchmark Audit

Debobrato Biswas and Imran Mahmud

Abstract—Counterfeit high-value banknotes harm retailers and individuals who cannot afford dedicated verification hardware. This paper presents a smartphone-based authentication system for Bangladeshi Tk 500 and Tk 1000 notes and evaluates it on JaalTaka, a public benchmark of 1,390 notes (802 genuine and 588 counterfeit notes seized by law enforcement) with six photographs per note. The system fine-tunes an ImageNet-pretrained EfficientNet-B0, fuses the predictions of up to six photographs of the same note, and returns a three-way verdict (genuine, uncertain, or counterfeit) with an adjustable strictness threshold. A Mahalanobis-plus-residual test in the network's embedding space rejects photographs that do not resemble the supported notes, and Grad-CAM heatmaps show which regions drove each decision. On a held-out test set of 208 notes that share no note with training, the system reaches 98.08% note-level accuracy (95% CI 95.16–99.25%) and an ROC-AUC of 0.994. It runs at 37 ms per photograph on a desktop CPU. An audit of the benchmark shows why these figures must be read with care. Pixel-level matching groups the 1,390 notes into only 846 distinct photograph sets: 116 of the 208 test notes have an exact or augmented copy in the training or validation split, and 12 duplicate groups carry both labels. All four test errors fall in these label-conflict groups. On the 92 test notes without any copy in training, the system makes no errors, but so do a 1-nearest-neighbour classifier and a linear probe on frozen ImageNet features. The benchmark is therefore saturated under note-level splitting. We release the duplicate groups and recommend group-aware splits and counterfeit-family hold-out tests for future work.

Index Terms—Banknote authentication, Bangladeshi Taka, counterfeit detection, data leakage, EfficientNet, explainable AI, Grad-CAM, out-of-distribution detection, transfer learning.

I. Introduction

COUNTERFEIT banknotes undermine trust in cash, which remains the main means of payment for small retailers and much of the population in Bangladesh. The Tk 500 and Tk 1000 notes are the preferred targets of counterfeiters because a single accepted forgery causes a large loss. Banks use sorting machines with ultraviolet, infrared, and magnetic sensors. Shopkeepers, transport workers, and ordinary citizens, however, rely on visual and tactile checks of the watermark, security thread, and intaglio print, which are slow and unreliable for untrained users. A smartphone camera is the one sensor almost everyone carries, which makes image-based authentication on commodity devices an attractive complement to manual checks.

Earlier image-based counterfeit detectors relied on hand-crafted features such as luminance histograms [1], background printing texture [2], or specific security features [3], often under controlled scanner or multispectral imaging [4]. Convolutional neural networks (CNNs) trained on smartphone photographs have since been shown to detect forgeries of several currencies [5], [6]. For Bangladeshi currency, most published work addressed denomination recognition [7], [8]. The recent JaalTaka dataset [9] made counterfeit detection research possible by releasing smartphone photographs of genuine notes and of counterfeits seized by the Rapid Action Battalion (RAB). Its authors reported test accuracies of 97.4–98.9% for five ImageNet CNNs.

High accuracy on a benchmark is necessary, but not sufficient, for a tool that people will trust with money. A deployable system must also (i) behave sensibly when the input is not a supported note, (ii) admit uncertainty instead of forcing a binary answer, (iii) explain its decision, and (iv) be evaluated in a way that predicts its behaviour on notes it has never seen. The last point is easy to get wrong. Deep networks readily exploit shortcuts [10], and duplicated or near-duplicated samples across training and test splits silently inflate reported accuracy, a problem that has affected many applied machine-learning studies [11].

This paper addresses these requirements. Its contributions are as follows.

1) A complete, open, CPU-real-time authentication pipeline for Tk 500 and Tk 1000 notes (Fig. 2). It combines an EfficientNet-B0 [12] classifier, fusion of up to six photographs of one note, a three-way decision rule with an adjustable strictness threshold, an embedding-space out-of-distribution (OOD) gate, and Grad-CAM [13] explanations.
2) A note-disjoint evaluation on JaalTaka that reports confidence intervals, per-photograph and per-note results, threshold behaviour, OOD rejection rates, and comparisons with frozen-feature baselines.

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3) An audit of the benchmark using pixel-level duplicate matching. It reveals that most test notes have copies in the training split and that several duplicate groups carry contradictory labels. It also shows that every test error of our system is caused by such a label conflict, and that the benchmark is saturated once duplicates are controlled. An occlusion test further indicates that part of the discriminative signal lies outside the note's central region.

The remainder of the paper is organised as follows. Section II reviews related work. Section III describes the dataset and protocol, and Section IV the proposed system. Section V reports the experiments and the benchmark audit. Section VI discusses limitations, and Section VII concludes the paper.

II. Related Work

A. Hand-Crafted Counterfeit Detection

Early systems extracted engineered features from images captured under controlled conditions. Yeh et al. [1] classified luminance histograms of Taiwanese banknotes with multiple-kernel support vector machines (SVMs). Berenguel et al. [2] analysed the fine background printing texture captured by a flatbed scanner, which differs between the intaglio and offset printing of genuine notes and the inkjet or laser printing of most forgeries. Baek et al. [4] exploited multispectral images, in which genuine inks respond differently from counterfeit ones. For Bangladeshi currency, Uddin et al. [3] inspected individual security features in banknote images. Dittimi and Suen [14] brought counterfeit detection to a mobile application. These methods are interpretable and light, but they depend on controlled imaging and on carefully tuned features for each currency and denomination.

B. Deep Learning for Banknotes

Transfer learning from ImageNet [15], [16] allows CNNs to be trained with the few thousand images that are typically available for banknotes. Pham et al. detected fake banknotes from visible-light smartphone images to assist visually impaired users [5] and later studied the cross-dataset setting for multinational banknotes [6]. For Bangladeshi notes, Jahangir and Chowdhury [7] used a neural network for denomination recognition, and Linkon et al. [8] combined lightweight CNNs with transfer learning for denomination detection. Counterfeit detection on the JaalTaka dataset was first benchmarked by its authors [9] with DenseNet121 [17], ResNet50 [18], VGG16 [19], MobileNetV2 [20], and InceptionV3 [21], and transfer learning has also been applied to counterfeit Bangladeshi notes in [22]. Table I summarises these studies. None of them evaluates OOD inputs, uncertainty, or the duplicate structure of the data.

C. Reliability: OOD Detection, Explanation, and Leakage

A classifier trained on two classes assigns one of them to any input, including photographs of other denominations or unrelated objects. Hendrycks and Gimpel [23] showed that the maximum softmax probability is a useful but weak OOD signal. Lee et al. [24] obtained stronger detection by fitting Gaussian models to intermediate features and scoring inputs by their Mahalanobis distance, an approach we adapt to a low-dimensional subspace. Neural network probabilities are also often miscalibrated [25], which motivates an explicit "uncertain" outcome instead of a single fixed threshold. Grad-CAM [13] highlights the image regions that support a class and is widely used to check what a CNN has learned. Finally, Geirhos et al. [10] described how networks solve benchmarks through unintended shortcuts, and Kapoor and Narayanan [11] documented how leakage between training and test data, including duplicated samples, inflates published results. Our audit in Section V-H applies these lessons to counterfeit banknote detection.

III. Dataset and Evaluation Protocol

A. The JaalTaka Dataset

JaalTaka [9], [26] contains photographs of 1,390 Bangladeshi notes: 802 genuine notes (398 Tk 500 and 404 Tk 1000) collected from three national banks, and 588 counterfeit notes (320 Tk 500 and 268 Tk 1000) supplied by RAB for research. Each note was photographed six times with one of four smartphones (Samsung Galaxy A50 and M31, Redmi Note 8 and Note 9 Pro), giving 8,340 images. The six photographs cover different segments of the front and back of the note (Fig. 1). The dataset description states that 649 original notes (502 genuine and 147 counterfeit) were expanded to 1,390 by augmentation. We resized every image so that its longer side is 1,024 pixels; the data are distributed under the CC BY 4.0 licence.

B. Note-Disjoint Split

Because the six photographs of a note are strongly correlated, we split the data by note, never by photograph. Within each class, notes were shuffled with a fixed seed (42) and assigned 70%/15%/15% to training, validation, and test, so that no note contributes photographs to more than one split. Table II lists the result. The validation split is used for model selection and to calibrate the OOD gate; the test split is used once, for the results in Section V. As Section V-H shows, a note-disjoint split is not sufficient on this dataset, because distinct note identifiers can share the same photographs.

IV. Proposed System

Fig. 2 shows the processing chain. A user photographs one note one to six times. Each photograph is classified independently. Photographs that fail the OOD gate are discarded, the remaining probabilities are averaged, and a three-way decision is made. A Grad-CAM heatmap accompanies every photograph.

TABLE I
Representative Banknote Authentication Studies

Study	Currency	Imaging	Approach and task
Yeh et al. [1] (2011)	Taiwan dollar	Image sensor	Luminance histograms + multiple-kernel SVM; counterfeit detection
Berenguel et al. [2] (2016)	Banknotes	Flatbed scanner	Background printing-texture features + SVM; counterfeit detection
Uddin et al. [3] (2016)	BDT	Camera images	Image-based analysis of security features; counterfeit detection
Baek et al. [4] (2018)	Banknotes	Multispectral	Spectral response of inks; counterfeit detection
Dittimi and Suen [14] (2018)	Banknotes	Smartphone	Mobile application; counterfeit detection
Pham et al. [5] (2020)	Banknotes	Smartphone, visible light	CNN; fake-banknote detection for visually impaired users
Linkon et al. [8] (2020)	BDT	Camera images	Lightweight CNN + transfer learning; denomination recognition
Pham et al. [6] (2022)	USD, EUR, KRW, JOD	Smartphone	CNN; fake detection across datasets
Piyas et al. [9] (2025)	BDT 500/1000	Smartphone (4 models)	JaalTaka dataset; five ImageNet CNN baselines (best 98.9%)
Tabassum et al. [22] (2025)	BDT	Images	Transfer learning; counterfeit detection
This work	BDT 500/1000	Smartphone, 1–6 views	EfficientNet-B0, multi-view fusion, three-way decision, OOD gate, Grad-CAM, duplicate audit

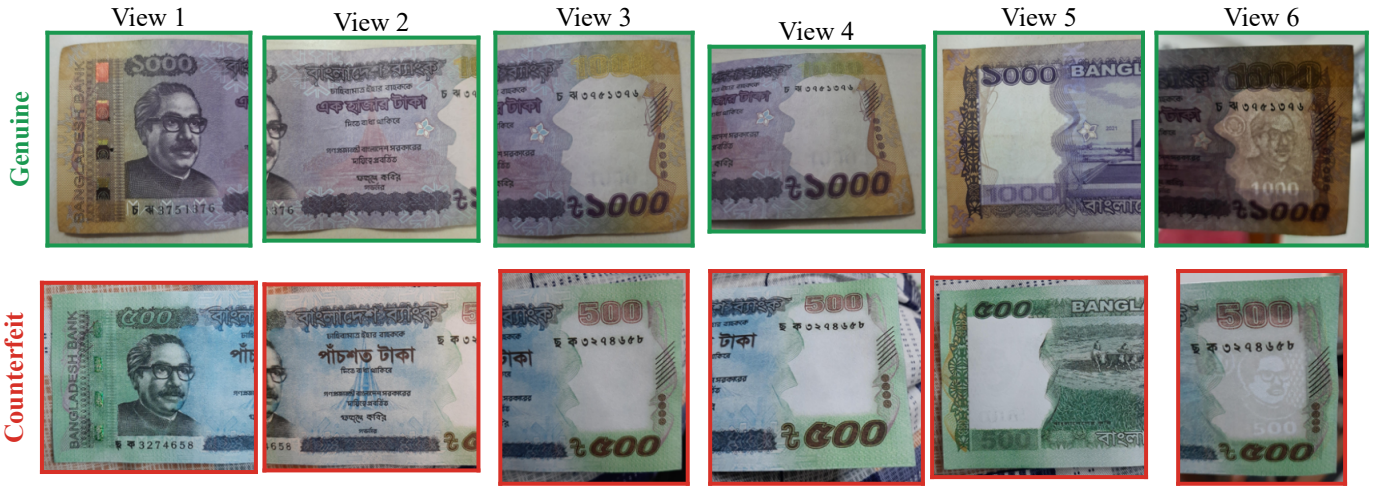


Fig. 1. The six photographs of one genuine Tk 1000 note (top) and one counterfeit Tk 500 note (bottom) from the JaalTaka test split. Each view shows a different segment of the note; the colour, lighting, and background vary with the phone and the session.

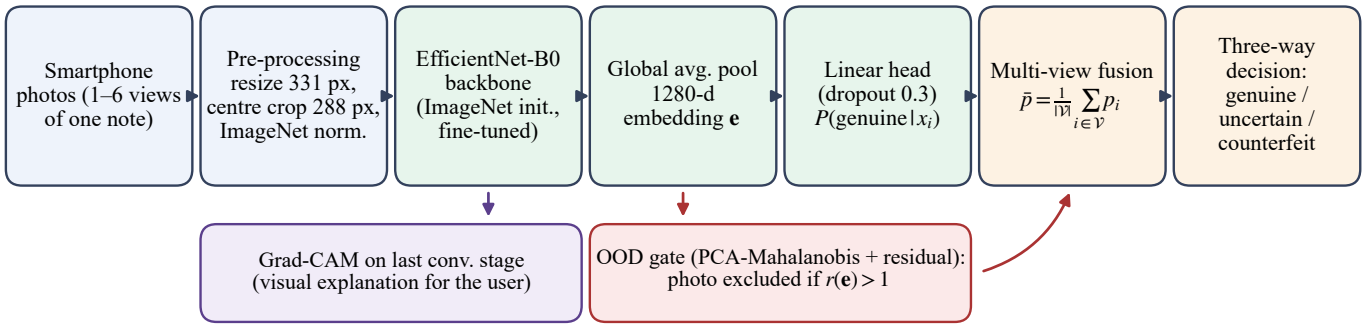


Fig. 2. Overview of the proposed system. Each photograph x_i of a note is classified by a fine-tuned EfficientNet-B0. Photographs whose embedding $\mathbf{e}$ fails the OOD test are excluded, the genuine probabilities of the remaining views are averaged, and the note is labelled genuine, uncertain, or counterfeit. Grad-CAM explains each per-photograph decision.

A. Classifier

We use EfficientNet-B0 [12], which offers a good trade-off between accuracy and computation on CPUs and mobile devices, initialised with ImageNet weights [15]. The convolutional feature extractor $f(\cdot)$ maps an input x to a $1280 \times 9 \times 9$ feature map for 288×288 inputs; global

TABLE II
Note-Disjoint Split of JaalTaka (Seed 42)

	Notes			Photographs	
Split	Genuine	Counterf.	Total	Genuine	Counterf.
Training	562	412	974	3,372	2,472
Validation	120	88	208	720	528
Test	120	88	208	720	528
Total	802	588	1,390	4,812	3,528

average pooling yields the embedding $\mathbf{e} = f(x) \in \mathbb{R}^{1280}$. The ImageNet classifier is replaced by dropout ($p = 0.3$) and a linear layer with two outputs, so that

$$p(x) = \mathrm{softmax}(\mathbf{We} + \mathbf{b})_{\text{genuine}}\,. \tag{1}$$

The model has 4.01 M parameters.

B. Training

Training images are augmented with random resized crops (scale 0.35–1.0, aspect ratio 0.75–1.33, output 288×288), rotations of up to $\pm 10°$, colour jitter (brightness and contrast ± 0.4, saturation ± 0.3, hue ± 0.04), and, with probability 0.2, Gaussian blur. The augmentations encourage the network to rely on print and texture rather than lighting, framing, or background. At evaluation, the shorter image side is resized to 331 pixels and the central 288×288 region is kept, which removes most of the background.

The first four stages of EfficientNet-B0 (the stem and the first three MBConv stages), which compute generic edge and colour filters, are frozen, and their batch-normalisation statistics are kept from ImageNet. The rest of the network is fine-tuned with the class-weighted, label-smoothed [21] cross-entropy

$$\mathcal{L} = -\sum_{c \in \{0,1\}} w_c\, q_c \log \hat{y}_c, \quad q_c = (1-\varepsilon)\mathbb{1}[c = y] + \tfrac{\varepsilon}{2}, \tag{2}$$

where $\hat{y}$ is the softmax output, $\varepsilon = 0.05$, and $w_c = N/(2N_c)$ compensates for the 562:412 class imbalance. We use AdamW [27] and a one-cycle learning-rate schedule [28]; Table III lists the settings. After each epoch, the model is evaluated on the validation split, and the weights with the highest sum of validation accuracy and AUC are kept. Validation accuracy rose from 98.80% after the first epoch to 99.04%, and validation AUC from 0.9919 to 0.9976 after the sixth, which was selected. Training took about 55 min on an Intel Core i5-12400 CPU without a GPU, using PyTorch [29].

C. Multi-View Fusion and Three-Way Decision

Let $\mathcal{V}$ be the set of photographs of one note that pass the OOD gate (Section IV-D). The note-level genuine probability is the mean

$$\bar{p} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} p(x_i). \tag{3}$$

TABLE III
Training Configuration

Setting	Value
Backbone	EfficientNet-B0, ImageNet-1k weights
Frozen layers	Stem + first 3 MBConv stages
Head	Dropout 0.3, linear $1280 \to 2$
Input size	288×288
Optimiser	AdamW, weight decay 10^{-4}
Learning rate	One-cycle, peak 3×10^{-4}, 15% warm-up
Batch size / epochs	12 / 6 (487 steps per epoch)
Loss	Weighted cross-entropy, label smoothing 0.05
Model selection	Max. validation (accuracy + AUC)
Random seed	42 (split, initialisation, shuffling)

Averaging is simple, needs no extra training, and accepts any number of views, which suits an interactive application in which a user may submit one quick photograph or six careful ones; it is the late-fusion analogue of view pooling in multi-view CNNs [30]. Given a strictness threshold $\tau \in [0.5, 1)$, the verdict is

$$\text{verdict} = \begin{cases} \text{not supported} & \text{if } \mathcal{V} = \emptyset, \\ \text{genuine} & \text{if } \bar{p} \geq \tau, \\ \text{counterfeit} & \text{if } \bar{p} < 0.5, \\ \text{uncertain} & \text{otherwise.} \end{cases} \tag{4}$$

The asymmetric rule reflects the asymmetric cost of errors: accepting a counterfeit causes a direct loss, whereas an "uncertain" verdict only asks the user to check the watermark, security thread, and raised print by hand or to add photographs. The default is $\tau = 0.7$; users can choose 0.6, 0.7, or 0.85, or set any value in the interface.

D. Out-of-Distribution Gate

Following the feature-space approach of Lee et al. [24], we model the embeddings of the training photographs. Let $\boldsymbol{\mu}$ be their mean and $\mathbf{U}_k \in \mathbb{R}^{1280 \times k}$ the first $k = 64$ principal directions, with variances $\sigma_1^2, \ldots, \sigma_k^2$. For an embedding $\mathbf{e}$ with $\mathbf{z} = \mathbf{U}_k^\top(\mathbf{e} - \boldsymbol{\mu})$, we compute the Mahalanobis distance inside the subspace and the energy outside it:

$$m(\mathbf{e}) = \sum_{j=1}^{k} \frac{z_j^2}{\sigma_j^2}, \qquad \rho(\mathbf{e}) = \|(\mathbf{e} - \boldsymbol{\mu}) - \mathbf{U}_k \mathbf{z}\|_2^2\,. \tag{5}$$

Both thresholds, θ_m and θ_ρ, are set to the 99.5th percentile of the validation photographs, and the OOD ratio is $r(\mathbf{e}) = \max\{m(\mathbf{e})/\theta_m,\ \rho(\mathbf{e})/\theta_\rho\}$. A photograph is accepted when $r(\mathbf{e}) \leq 1$ and when its texture, measured as the variance of the Laplacian of a 512×512 grey-scale version, is at least 30. The texture test rejects blank, heavily blurred, and featureless images. The residual term catches inputs whose embedding leaves the subspace spanned by banknote photographs, which the in-subspace Mahalanobis distance alone cannot detect.

Algorithm 1 Authentication of One Banknote

Require: Photographs $x_1, \ldots, x_n$ of one note ($1 \le n \le 6$), threshold τ
Ensure: Verdict, $\bar{p}$, and one heatmap per photograph
$\mathcal{V} \leftarrow \emptyset$
for $i = 1$ to n **do**
  $\mathbf{e}_i \leftarrow f(\text{crop}(x_i))$; $p_i \leftarrow$ Eq. (1)
  $H_i \leftarrow$ Grad-CAM of the predicted class, Eq. (6)
  if $r(\mathbf{e}_i) \le 1$ and $\text{texture}(x_i) \ge 30$ **then**
    $\mathcal{V} \leftarrow \mathcal{V} \cup \{i\}$
  end if
end for
if $\mathcal{V} = \emptyset$ **then**
  return not supported
end if
$\bar{p} \leftarrow \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} p_i$
return verdict from Eq. (4), $\bar{p}$, $\{H_i\}$

E. Visual Explanation

For each photograph, Grad-CAM [13] is computed on the last convolutional feature map $A \in \mathbb{R}^{1280 \times 9 \times 9}$ for the predicted class c:

$$\alpha_k = \frac{1}{81} \sum_{u,v} \frac{\partial y^c}{\partial A^k_{uv}}, \qquad L^c = \text{ReLU}\Big(\sum_k \alpha_k A^k \Big), \tag{6}$$

where y^c is the logit of class c. The map is normalised, upsampled to the analysed region, and overlaid on the photograph.

F. Inference Procedure and Deployment

Algorithm 1 summarises inference. The deployed detector averages two crops per photograph (the central crop of Section IV-B and the full frame resized to 288×288); all results in Section V use the central crop only, which matches the training-time evaluation. The model runs in 37 ms per photograph (median of 30 runs, batch size one, six threads of the i5-12400 CPU). The system is packaged as two local web applications: a Flask interface and a Streamlit interface that show the verdict, the per-photograph probabilities, the Grad-CAM heatmaps, and the effect of changing τ.

V. Experiments and Results

A. Metrics

Genuine is the positive class. We report accuracy; the counterfeit detection rate (specificity, the share of counterfeits rejected); the genuine acceptance rate (recall); and the area under the ROC curve (AUC) [31]. Accuracy is given with a 95% Wilson score interval [32]. Results are reported per photograph (n = 1,248) and per note after fusing all six photographs (n = 208). Unless stated otherwise, decisions use the threshold 0.5, i.e., the two-way limit of Eq. (4).

TABLE IV
Test Results of the Proposed System (Threshold 0.5)

Metric	Per photograph	Per note (6 views)
Samples (genuine / counterf.)	1,248 (720 / 528)	208 (120 / 88)
Accuracy (%)	98.08	98.08
95% Wilson CI (%)	97.15–98.70	95.16–99.25
Counterfeit detection (%)	97.73	97.73 (86/88)
Genuine acceptance (%)	98.33	98.33 (118/120)
Balanced accuracy (%)	98.03	98.03
F1, genuine class	0.983	0.983
ROC-AUC	0.990	0.994

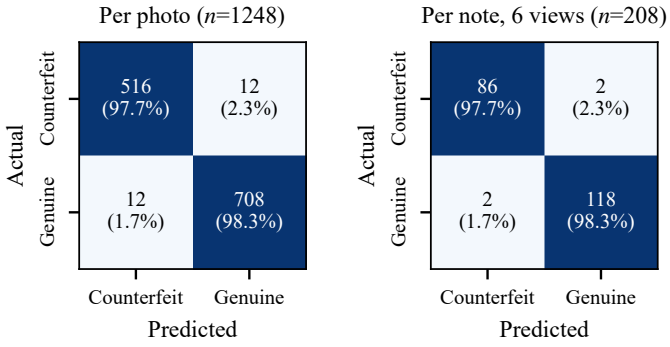


Fig. 3. Confusion matrices on the test split, per photograph (left) and per note after fusing six photographs (right). Percentages are row-normalised.

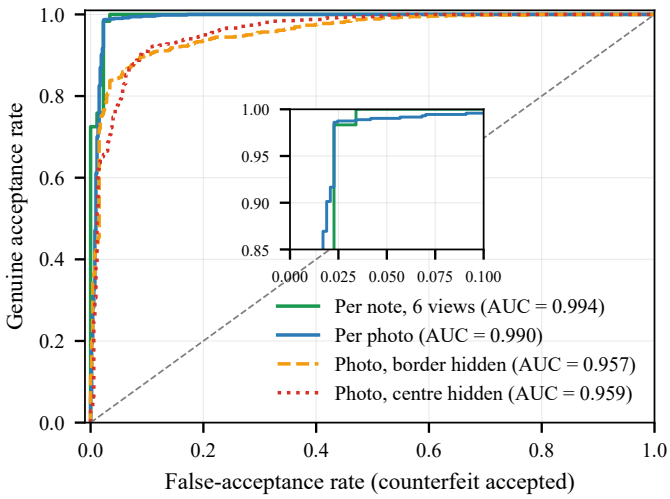


Fig. 4. ROC curves on the test split for per-note fusion, single photographs, and the two occlusion diagnostics of Section V-G. The inset magnifies the low-false-acceptance region.

B. Main Results

Table IV and Fig. 3 summarise the test results. Per photograph, the system reaches 98.08% accuracy and an AUC of 0.990. Per note, it rejects 86 of 88 counterfeits and accepts 118 of 120 genuine notes (98.08% accuracy, AUC 0.994). Fig. 4 shows that both ROC curves rise almost vertically; at the default operating point, 2.3% of counterfeit notes are accepted.

The per-note confusion matrix is exactly the per-photograph matrix divided by six: every error occurs on all six photographs of the same note. The same

TABLE V
Comparison With Frozen-Feature Baselines on the Test Split

	Per photograph		Per note	
Method	Acc. (%)	AUC	Acc. (%)	AUC
1-NN, frozen ImageNet features	96.71	0.967	97.12	0.979
Linear probe, frozen features	96.47	0.981	98.08	0.983
Fine-tuned EfficientNet-B0 (ours)	98.08	0.990	98.08	0.994

TABLE VI
Effect of the Number of Fused Photographs (Test Notes)

Photographs k	1	2	3	4	5	6
Subsets per note	6	15	20	15	6	1
Accuracy (%)	98.08	98.08	98.08	98.08	98.08	98.08
Mean AUC	0.9900	0.9912	0.9917	0.9925	0.9930	0.9939

holds for each view position separately (all six reach 98.08%). The errors are therefore not caused by an unlucky photograph or viewing angle, but by four specific notes, which Section V-H examines.

C. Comparison With Baselines

To judge how much of the performance is due to fine-tuning, we evaluated two classifiers on frozen ImageNet EfficientNet-B0 embeddings with the same split and the same evaluation crop: a 1-nearest-neighbour (1-NN) classifier with cosine similarity, and a class-balanced logistic-regression linear probe. Note-level scores average the per-photograph outputs, as in Eq. (3). Table V shows that fine-tuning improves per-photograph accuracy by 1.4–1.6 points and AUC by 0.009–0.023. At note level, however, the linear probe matches the accuracy of the fine-tuned model, and even the 1-NN classifier errs on only six notes. For reference, the JaalTaka authors reported 97.4–98.9% for five fine-tuned CNNs [9]; their split protocol differs from ours, so those figures are not directly comparable.

D. Multi-View Fusion and the Strictness Threshold

To measure the value of additional photographs, we evaluated Eq. (3) on every subset of k views of each test note ($\binom{6}{k}$ subsets per k). Table VI shows that accuracy is the same for every subset, again because all errors are note-level, whereas the mean AUC increases monotonically from 0.990 with one photograph to 0.994 with six: fusion makes the ranking of notes more reliable and pushes the scores of both classes further from the decision boundary.

Fig. 5(a) shows that the fused scores are strongly bimodal: all correctly classified genuine notes have $\bar{p} > 0.86$ and all correctly classified counterfeits $\bar{p} < 0.10$. Consequently, the three-way rule of Eq. (4) leaves the outcome unchanged for $\tau \leq 0.85$; at $\tau = 0.90$, one genuine note (0.8%) becomes "uncertain" (Fig. 5(b)). The two accepted counterfeits have $\bar{p} \approx 0.95$ and would only be caught with $\tau > 0.954$, at which point most genuine notes would also be sent to manual inspection. The threshold therefore cannot fix these two errors, which is consistent with the finding of Section V-H that they are not ordinary misclassifications.

E. Out-of-Distribution Gate

Fig. 6 shows the distribution of the OOD ratio $r(\mathbf{e})$. By construction, about 0.5–1% of in-distribution photographs exceed the limit: on the test split, 4 of 1,248 photographs (0.32%) are rejected, against 0.72% on the validation split used for calibration. The median ratio of test photographs is 0.26, far below the limit. Because a note is analysed whenever at least one of its photographs passes, no test note was declared "not supported". In a preliminary check, the gate rejected 100 of 105 photographs of other Bangladeshi denominations (Tk 2–200) taken from a separate collection, as well as blank, noise, and unrelated images. A systematic OOD benchmark with independently collected images remains future work.

F. Visual Explanations

Fig. 7 shows Grad-CAM heatmaps for the genuine test note with the highest fused score, the counterfeit note with the lowest, and one of the two accepted counterfeits. For correct decisions, the heatmaps concentrate on the printed portrait and on the denomination text and ornament panels, regions that carry the finest printed detail of the note. On the counterfeit in row (b), the left border with the vertical security band also contributes. The heatmaps of the wrongly accepted note in row (c) look like those of a genuine note, which is expected if, as shown next, an identical copy of this note appears in the training set labelled as genuine.

G. Shortcut Diagnostic

To test whether the model relies on the note or on its surroundings, we blacked out either the central region of each test input (keeping a 20% border) or the border (keeping the centre), as illustrated in Fig. 8, and evaluated the unchanged model. Hiding the border reduces accuracy to 88.94% (AUC 0.957). Hiding the centre reduces accuracy to 75.48%, because most genuine photographs are then rejected (genuine acceptance 58.47%), but the AUC remains 0.959 and 98.67% of counterfeits are still recognised. Information sufficient to rank most photographs correctly is therefore present in the 20% border alone. Part of the border is the note itself (margins, security band, serial number), but in many photographs it also contains the table or hand on which the note lies, and the genuine and counterfeit notes came from different sources (banks and RAB) and were likely photographed in different sessions. We cannot rule out that capture conditions such as background, lighting, or the phone used contribute to the decision, a classic shortcut risk [10].

H. Benchmark Audit: Duplicates and Label Conflicts

The near-perfect agreement between all photographs of a note, the strength of the non-adaptive baselines,

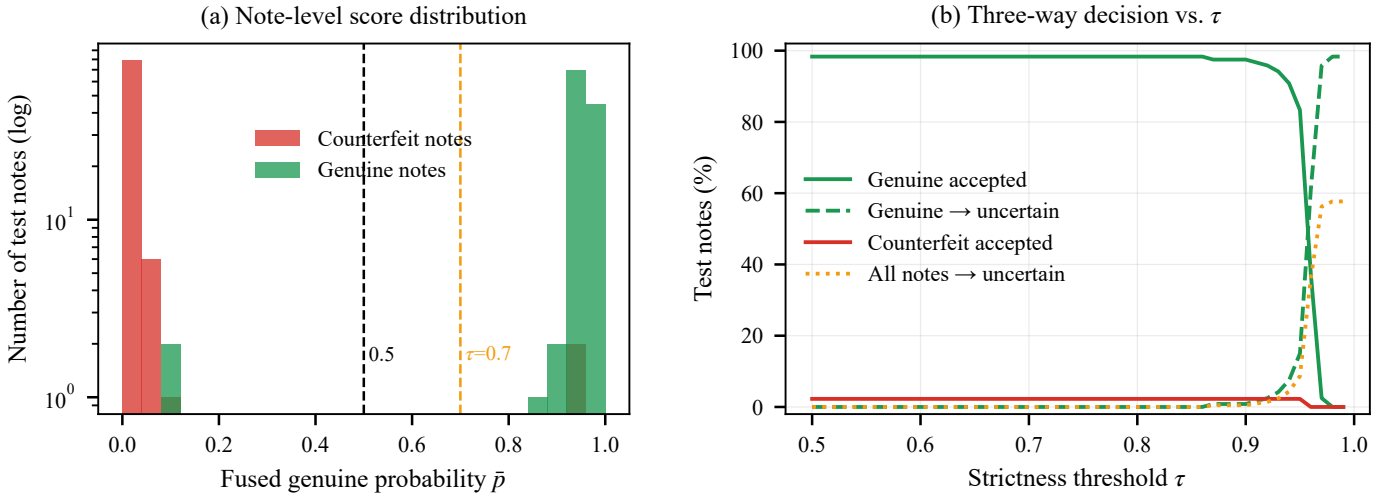


Fig. 5. (a) Distribution of the fused genuine probability $\bar{p}$ of the 208 test notes (log scale); dashed lines mark the counterfeit limit 0.5 and the default strictness $\tau = 0.7$. (b) Share of test notes accepted as genuine, sent to "uncertain", or wrongly accepted, as a function of the strictness threshold τ.

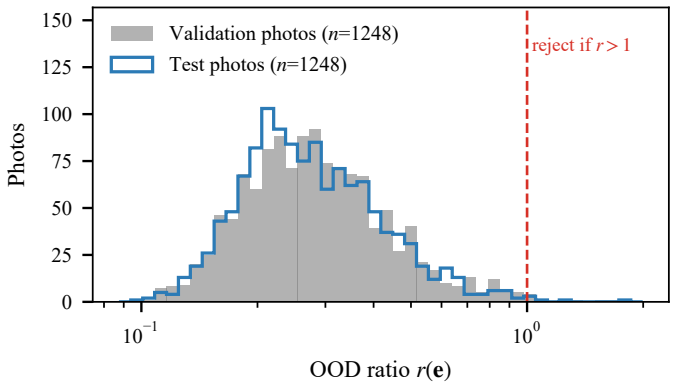


Fig. 6. OOD ratio $r(\mathbf{e})$ of validation photographs (used for calibration) and test photographs. Photographs to the right of the dashed line are excluded from the note-level decision.

TABLE VII
Duplicate Audit of JaalTaka (Pixel NCC > 0.98 Over Six Views)

Quantity	Value
Notes / distinct photograph sets (groups)	1,390 / 846
Groups of size 1 / 2 / 3	449 / 250 / 147
Notes sharing their photographs with another note	941 (67.7%)
Groups with only counterfeit / only genuine labels	289 / 545
Groups containing both labels	12 (31 notes)
Test notes with a copy in training or validation	116 of 208 (55.8%)
counterfeit / genuine	60 of 88 / 56 of 120
Test notes in label-conflict groups	6
Test errors explained by label conflicts	4 of 4

and the statement that the dataset was enlarged by augmentation led us to search for duplicated notes. For every pair of notes, we computed the normalised cross-correlation (NCC) between 48×48 grey-scale thumbnails of corresponding views, which is invariant to brightness and contrast changes, averaged it over the six views, and took the maximum over horizontal and vertical flips. The distribution of the best match is clearly bimodal (Fig. 9(a)): for the validation and test notes, matches below the gap reach at most 0.963, whereas matches above it are at least 0.997. We therefore grouped notes whose six views match with NCC > 0.98 (connected components). Table VII summarises the findings.

Three findings stand out. First, the 1,390 notes contain only 846 distinct photograph sets, so the augmented copies mentioned in the dataset description were stored as independent notes. A note-level split thus places copies of 55.8% of the test notes in the training or validation data. Second, 12 groups contain both genuine- and counterfeit-labelled copies of the same photographs (e.g., genuine notes 400–406 duplicate counterfeit notes 106–112 and 547–553, and counterfeit notes 400 and 403–406 duplicate genuine notes 802 and 1–4). At least one label in each of these groups must be wrong. The genuine-labelled notes 402 and 403, for instance, carry the serial number 1843834, which also appears on several counterfeit-labelled notes in the dataset. Third, all four test errors of our system are notes from these label-conflict groups, and in each case the model predicts the label of the copy it saw during training.

We therefore re-scored the system and the baselines on two subsets of the original test split (Table VIII). On the 92 clean test notes, whose duplicate group has no member in training or validation, all three methods are error-free. After removing only the six label-conflict notes, all three are also error-free on the remaining 202 notes. The reported 98.08% note-level accuracy is thus not limited by the model. It is the label noise of the benchmark, and on correctly labelled notes the task is saturated even for a nearest-neighbour classifier on generic features.

VI. Discussion

A. What the Results Do and Do Not Show

Within JaalTaka, the proposed system separates genuine and counterfeit notes perfectly once label conflicts are

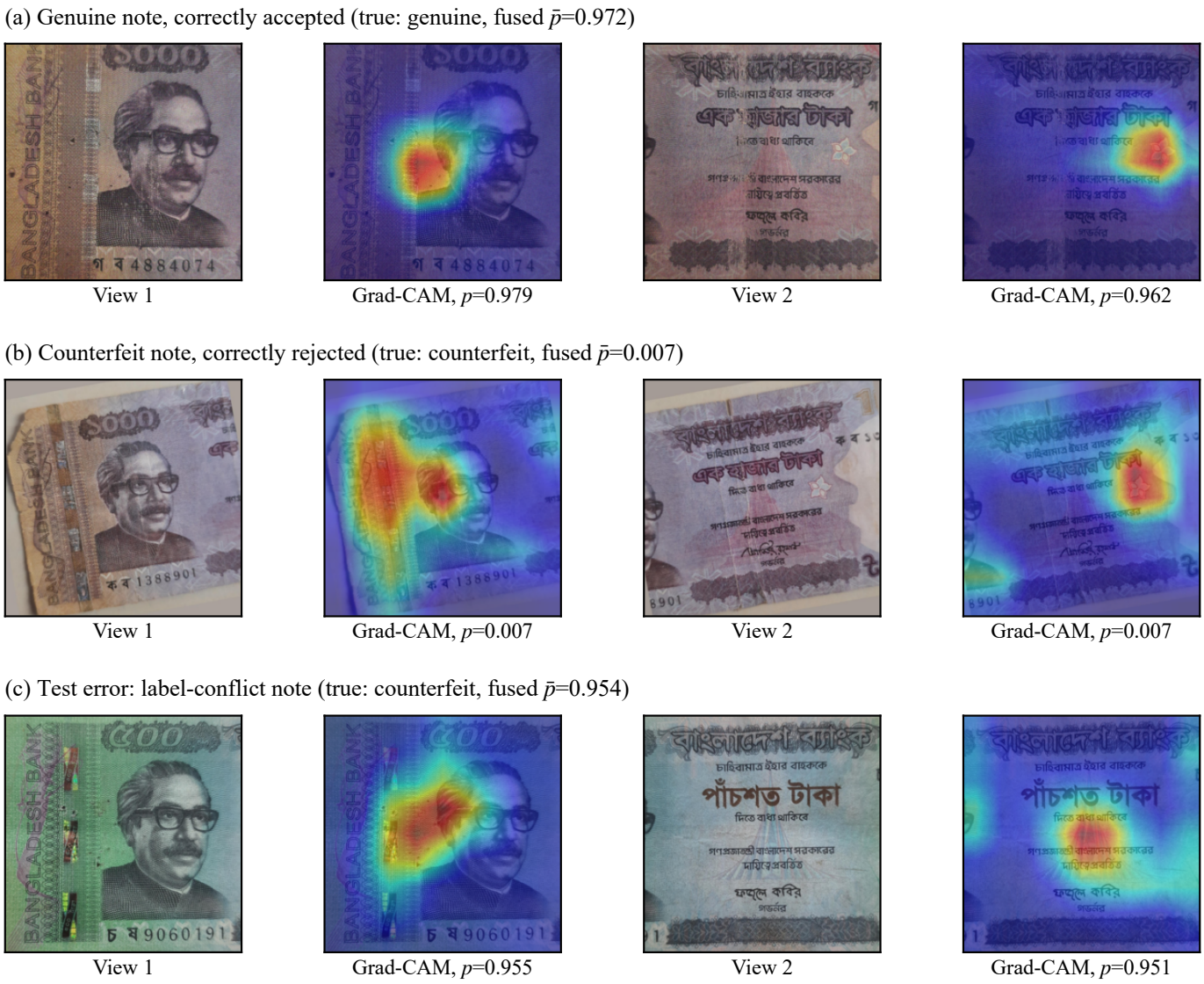


Fig. 7. Grad-CAM explanations on held-out test notes (views 1 and 2; red indicates high relevance). (a) Genuine note correctly accepted. (b) Counterfeit note correctly rejected. (c) Counterfeit note wrongly accepted; its photographs also occur in the training set under the genuine label (Section V-H).

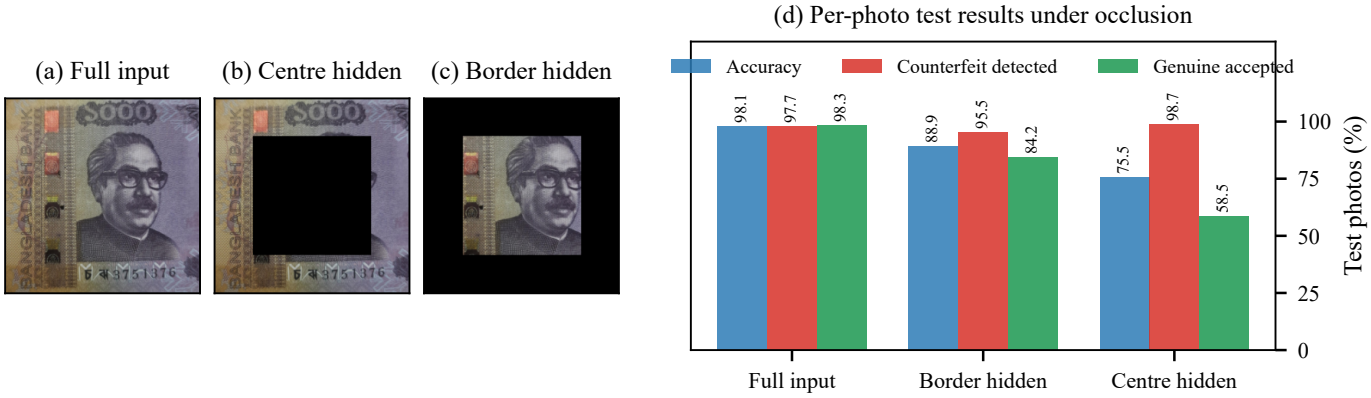


Fig. 8. Occlusion diagnostic. (a)–(c) Evaluation input, with the centre or the 20% border hidden. (d) Per-photograph test accuracy, counterfeit detection rate, and genuine acceptance rate of the unchanged model under each condition.

removed, with a lower 95% confidence bound of 96.0% on the 92 notes that have no copy in training. Its scores are

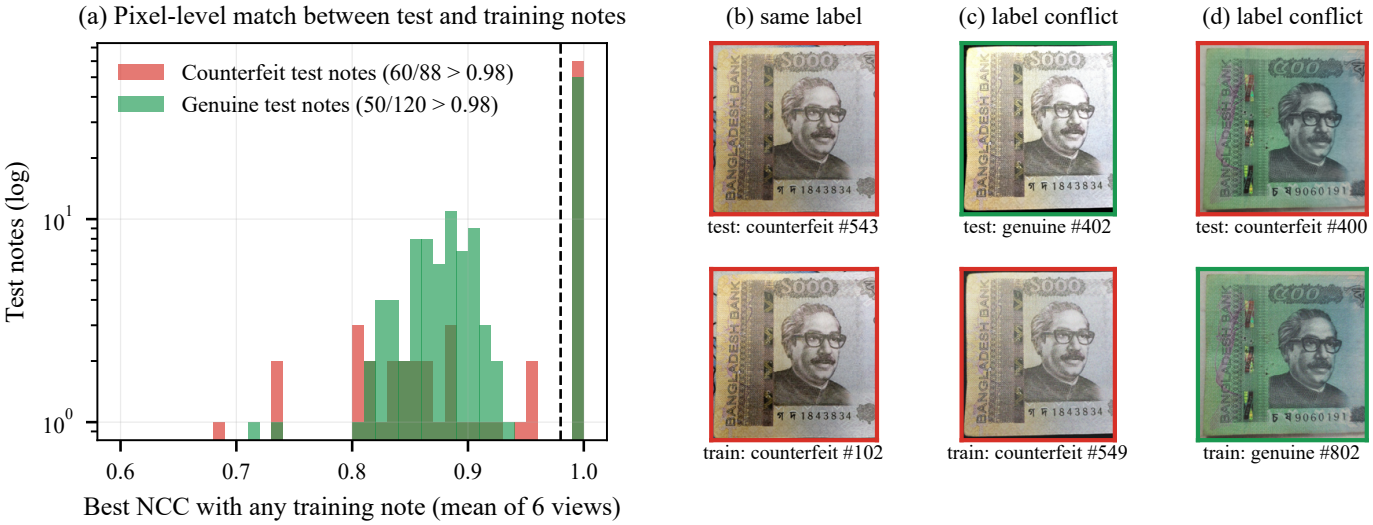


Fig. 9. Duplicate audit. (a) Best pixel-level match (mean NCC over the six views) between each test note and any training note; the dashed line marks the duplicate threshold 0.98. (b)–(d) First view of a test note (top) and its matching training note (bottom); frame colour gives the dataset label (green: genuine, red: counterfeit). (b) An augmented copy with the same label. (c), (d) Identical photographs stored under opposite labels.

TABLE VIII
Note-Level Accuracy (%) on Audited Test Subsets

Test subset	All	Without label conflicts	Clean (no copy)
Notes (genuine / counterf.)	208 (120/88)	202 (117/85)	92 (64/28)
1-NN, frozen features	97.12	100.0	100.0
Linear probe, frozen	98.08	100.0	100.0
Ours (fine-tuned)	98.08	100.0	100.0
95% Wilson CI	95.16–99.25	98.13–100	95.99–100

well separated, its errors are fully explained, and it runs in real time on a CPU. These results do not, however, show that the system will detect new counterfeits. The 588 counterfeit notes collapse into 289 counterfeit-only photograph sets, the counterfeits we inspected visually share a small number of serial numbers, and they were collected separately from the genuine notes. A model, or even a 1-NN classifier, can therefore succeed by recognising the visual signature of known counterfeit batches or capture sessions, as the occlusion diagnostic suggests, rather than generic evidence of forgery. Saturation also means that the benchmark can no longer rank methods: the differences between published CNNs on JaalTaka are smaller than the effect of its label noise.

B. Recommendations for the Benchmark

For future work on JaalTaka and similar datasets, we recommend (i) removing or relabelling the 12 label-conflict groups; (ii) splitting by duplicate group, not by note identifier, for which we release our group list; (iii) holding out entire counterfeit families (e.g., by serial number or seizure batch) and capture devices, to measure generalisation to unseen forgeries and phones; and (iv) reporting results with confidence intervals and simple frozen-feature baselines, which immediately reveal a saturated benchmark.

C. Limitations

The system covers only Tk 500 and Tk 1000, the denominations for which counterfeit photographs are publicly available; other notes are answered with “not supported”. A visible-light photograph cannot capture ultraviolet features, magnetic ink, or the tactile relief of intaglio print, so the output should be treated as decision support and suspicious notes should be confirmed at a bank. Close-up phone photographs are the only image type in the training data, and flatbed scans or distant photographs of the whole note are mostly rejected by the OOD gate. The OOD evaluation on other denominations is preliminary, and the class probabilities have not been calibrated on an independent dataset [25]. Finally, we did not retrain the model on a group-aware split; doing so, together with a counterfeit-family hold-out, is the most important next step.

VII. Conclusion

We presented a reliability-oriented smartphone system for detecting counterfeit Bangladeshi Tk 500 and Tk 1000 notes. It combines a fine-tuned EfficientNet-B0, fusion of up to six photographs, a three-way decision with adjustable strictness, an embedding-space OOD gate, and Grad-CAM explanations, and it runs in real time on a CPU. On a note-disjoint JaalTaka test split, it achieves 98.08% note-level accuracy and an AUC of 0.994. A pixel-level audit showed that this figure mainly reflects the benchmark: most test notes have copies in the training data, 12 duplicate groups carry contradictory labels, and these conflicts explain every test error, while on duplicate-free notes even simple baselines are error-free. Progress in

counterfeit detection now depends less on new architectures than on evaluation data that separates counterfeit families, capture sessions, and devices, and on reliability mechanisms such as those proposed here.

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Author Contributions

D. Biswas: conceptualisation, software, experiments, formal analysis, and writing of the original draft. I. Mahmud: supervision, methodology review, and review and editing of the manuscript.

Conflict of Interest

The authors declare that they have no conflict of interest.

Generative AI Disclosure

The authors used an AI assistant to help write analysis and plotting scripts and to edit the manuscript text. The authors checked all code, results, and references and take full responsibility for the content of this paper.

Data and Code Availability

The JaalTaka dataset is publicly available from Mendeley Data under the CC BY 4.0 licence [26]. The trained model, the training and evaluation code, the scripts that reproduce every table and figure, and the list of duplicate groups found in the audit are available from the corresponding author and will be released in a public repository upon publication.

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